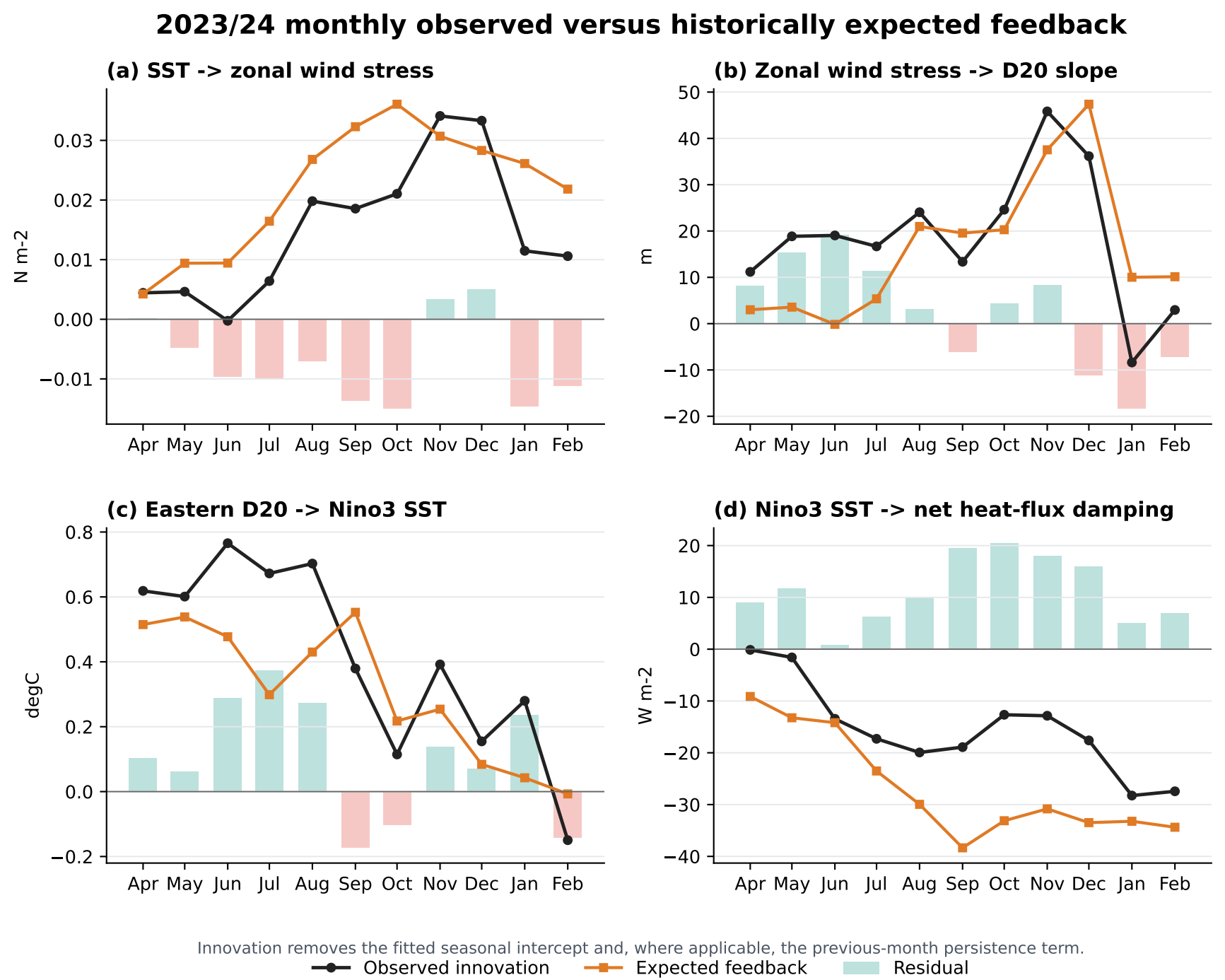
2023/24 monthly observed versus historically expected feedback
(a) SST -> zonal wind stress
N m-2
0.03
0.02
0.01
0.00
−0.01
Apr May Jun Jul Aug Sep Oct Nov Dec Jan Feb
(b) Zonal wind stress -> D20 slope
m
50
40
30
20
10
0
−10
−20
Apr May Jun Jul Aug Sep Oct Nov Dec Jan Feb
(c) Eastern D20 -> Nino3 SST
degC
0.8
0.6
0.4
0.2
0.0
−0.2
Apr May Jun Jul Aug Sep Oct Nov Dec Jan Feb
(d) Nino3 SST -> net heat-flux damping
W m-2
20
10
0
−10
−20
−30
−40
Apr May Jun Jul Aug Sep Oct Nov Dec Jan Feb
Innovation removes the fitted seasonal intercept and, where applicable, the previous-month persistence term.
Observed innovation
Expected feedback
Residual